Alejandro Sarasti Sierra

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Education

Universidad de los Andes — **Bogotá, Colombia**
M.A. in Economics (PEG), expected — *2025 – Dec 2026*
Thesis: *Real Effects of Accounting Changes: How Do Changes in Firms' Observables Affect Productivity and Access to Credit?*
Advisor: Christian Posso (Banco de la República).
GPA: 4.6/5.0

Universidad EIA — **Medellín, Colombia**
B.Sc. in Economics — *Dec 2024*
Thesis: *Valuing Data as an Intangible Asset.*
GPA: 4.2/5.0

Universidad de Antioquia — **Medellín, Colombia**
B.Eng. in Systems Engineering — *In progress*
Coursework completed; undergraduate thesis in progress: *Interpretability of Language Models: Identifying a Misalignment Direction in Internal Representations.*
GPA: 4.2/5.0

Research Experience

World Bank — **Medellín, Colombia**
Research Assistant — *Jan 2025 – Present*
Firm-level financial and administrative data for research on business dynamism and productivity in Colombia; productivity estimation in Stata and R.

Banco de la República (Central Bank of Colombia) — **Cartagena, Colombia**
Research Intern — *Jan – Dec 2024*
Center for Regional Economic Studies (CEER). Historical and contemporary demographic and economic data for research on subnational informality and demographic change; small area estimation and econometric models in R and Stata.

ECSIM — **Medellín, Colombia**
Research Assistant — *Jul 2021 – Oct 2022*
Implemented system-dynamics economic models in Matlab (Simulink).

Publications

Acosta, K., Jaramillo-Echeverri, J., Lasso-Jaramillo, D., & Sarasti-Sierra, A. (2025). **The geography of informality: the case of Colombia**. *Spatial Economic Analysis*. doi:10.1080/17421772.2025.2522807. Replication package: github.com/sarasti2/replication-geography-of-informality-colombia.

Ramírez-Giraldo, M. T. (coord.), Acosta-Ordoñez, K., . . . , Sarasti-Sierra, A., et al. (2026). **Explorando las brechas de género en Colombia** [Exploring gender gaps in Colombia]. *Ensayos sobre Política Económica (ESPE)*, No. 111. Banco de la República. doi:10.32468/espe111. Author of Section 1.1, *Gender demographic trends, 1973–2023*.

Otero-Cortés, A., Acosta-Ordoñez, K., . . . , Sarasti-Sierra, A., et al. (2025). **Nueva evidencia sobre la informalidad laboral y empresarial en Colombia** [New evidence on labour and business informality in Colombia]. *Ensayos sobre Política Económica (ESPE)*, No. 108. Banco de la República. doi:10.32468/espe108.

Work in Progress

Jaramillo-Echeverri, J., Sarasti-Sierra, A., & Flórez, L. A. **Gender Inequality and Demographic Change: Implications for Long-Run Growth**. Forthcoming in *Documentos de Trabajo sobre Economía Regional y Urbana*, Banco de la República.

Jaramillo-Echeverri, J., & Sarasti-Sierra, A. **Cambios demográficos departamentales a lo largo de los siglos XX y XXI** [Departmental demographic change across the 20th and 21st centuries]. Book chapter in an edited volume on 30 years of the Center for Regional Economic Studies (CEER), Banco de la República (eds. J. Bonet and C. Granger).

Presentations

Nov 2026: **Departmental demographic change across the 20th and 21st centuries**, with J. Jaramillo-Echeverri. Authors' workshop, volume on 30 years of the CEER, Banco de la República, Bogotá (25 November 2026, scheduled). Code: github.com/sarasti2/Demografia_30_CEER.

2026: **Gender Inequality and Demographic Change: Implications for Long-Run Growth**, with J. Jaramillo-Echeverri and L. A. Flórez.

Oct 2026: IX Congreso de Economía Colombiana [9th Colombian Economics Conference], Universidad de los Andes, Bogotá.

Apr 2026: III Congreso Nacional de la Asociación Colombiana de Demografía, Población y Desarrollo (POPULORUM 2026), Universidad Externado de Colombia, Bogotá. Earlier version, presented as *Brechas de género en las dinámicas demográficas*.

Teaching

Universidad de los Andes — **Continuing Education, School of Economics**
Co-instructor, Impact Evaluation with Machine Learning — *2027-1*
Upcoming course, with Ana María Patrón: causal machine learning for impact evaluation in R and Python.

Universidad de los Andes — **Bogotá, Colombia**
Teaching Assistant, Microeconometrics (graduate) — *Aug – Dec 2026*
Tutorial sessions, course materials and problem sets for the M.A. in Economics.

Additional Training

2023: **Micro-degree in Artificial Intelligence**, Fusemachines (288 hours). Computer vision and image processing with OpenCV, scikit-image and NumPy.

2021: **Summer School on the Economics of Climate Change**, Universidad EAFIT, Universidad EIA and Technical University of Munich (40 hours, plus a monograph on climate-smart agriculture in rural Colombia).

Awards and Honours

2022–2023: Dean's List (top academic performance), Universidad EIA: 2022-2, 2023-1, 2023-2.

Oct 2023: Second place, National Academic Economics Competition (CANE), Universidad del Rosario, Bogotá.

Sep 2023: EIA Datathon, Universidad EIA, Medellín.

Skills and Interests

Programming: **Advanced:** Python, Stata, R, LaTeX, Git. **Intermediate:** Matlab, Java, C++, Power BI, Vensim, ArcGIS, Docker.

Languages: Spanish (native), English (fluent, C1), Portuguese (basic, A2).

Research interests: Demographic dynamics; gender and development; growth and productivity; computational and econometric methods for scarce or imperfect data.

References

Manuel Fernández Sierra: Assistant Professor, Economics Department, Universidad de los Andes
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Mateo Uribe-Castro: Economics Department, Universidad de los Andes
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Juliana Jaramillo-Echeverri: Researcher, Center for Regional Economic Studies (CEER), Banco de la República
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